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Mathematics

Grade 7

Chapter: Comparing Quantities, Rational Numbers, Perimeter and Area, Algebraic Expressions, Exponents and Powers, Symmetry, Visualising Solid Shapes

Date: _____

Marks: 30

Time: 45 minutes

Multiple Choice Questions (MCQ)

1. Which of the following represents a **ratio**?
   (a) 3:4
   (b) 5%
   (c) 0.75
   (d) 2/5
   Answer:
2. What is 25 % of 200?
   (a) 25
   (b) 50
   (c) 75
   (d) 100
   Answer:
3. If a product is bought at **cost price** of ₹120 and sold at ₹138, the profit percent is?
   (a) 10 %
   (b) 15 %
   (c) 20 %
   (d) 25 %
   Answer:
4. Simplify the algebraic expression: **3x + 5x − 2x**.
   (a) 6x
   (b) 8x
   (c) 10x
   (d) 4x
   Answer:

5. Write 0.0045 in standard form.
   (a) 4.5×10^{-3}
   (b) 4.5×10^{-4}
   (c) 4.5×10^{-5}
   (d) 4.5×10^{-2}
   Answer:
6. Which of the following is a **rational number**?
   (a) $\sqrt{2}$
   (b) π
   (c) 7/3
   (d) $\sqrt{5}$
   Answer:
7. Evaluate: $(2^3)^2$.
   (a) 2^6
   (b) 2^5
   (c) 2^4
   (d) 2^3
   Answer:
8. The perimeter of a square with side **5 cm** is?
   (a) 10 cm
   (b) 20 cm
   (c) 25 cm
   (d) 40 cm
   Answer:
9. Find the area of a triangle with base 8 cm and height 5 cm.
   (a) 20 cm^2
   (b) 24 cm^2
   (c) 40 cm^2
   (d) 40 cm^2
   Answer:
10. Which solid has 6 faces, 12 edges, and 8 vertices?
   (a) Cube
   (b) Cylinder
   (c) Cone
   (d) Sphere

Answer:

11. If a discount of 20 % is given on a marked price of ₹250, the selling price is?
    (a) ₹200
    (b) ₹210
    (c) ₹225
    (d) ₹240
    Answer:
12. Convert 3/8 to a percentage.
    (a) 37.5 %
    (b) 40 %
    (c) 35 %
    (d) 42 %
    Answer:
13. Which of the following is true about **line symmetry**?
    (a) Rotation by 180°
    (b) Mirror image on opposite side
    (c) Only for circles
    (d) None of these
    Answer:
14. The expression **5y − 3y + 2y** simplifies to?
    (a) 4y
    (b) 6y
    (c) 2y
    (d) 8y
    Answer:
15. If $2^n = 32$, then n =?
    (a) 3
    (b) 4
    (c) 5
    (d) 6
    Answer:
16. Which of the following is NOT a **solid shape**?
    (a) Cylinder
    (b) Sphere

(c) Triangle

(d) Cone

Answer:

17. A rectangular garden is 12 m long and 8 m wide. Its **area** is?

    (a) 20 m^2

    (b) 40 m^2

    (c) 80 m^2

    (d) 96 m^2

    Answer:

18. Which of the following represents **exponential notation**?

    (a) 7 + 7 + 7

    (b) 7 × 7 × 7

    (c) 7^3

    (d) $7^2 + 7^1$

    Answer:

19. The ratio of boys to girls in a class is 3:2. If there are 15 boys, how many girls are there?

    (a) 10

    (b) 9

    (c) 8

    (d) 7

    Answer:

20. Which of the following is a **translation** in symmetry?

    (a) Flip over a line

    (b) Rotate 90°

    (c) Slide without turning

    (d) Mirror over diagonal

    Answer:

Fill in the Blanks

1. A number that can be written as **p/q** is called a **rational number**.

   Answer:

2. The **perimeter** of a square of side **s** is 4s.

   Answer:

3. The **exponent** in 2^3 tells how many times 2 is multiplied by itself.

Answer:

4. In the expression 5x − 3, 5 is a **coefficient** and x is a **variable**.

Answer:

5. A shape that is identical on both sides of a line has **line symmetry**.

Answer:

True or False

1. The sum of interior angles of a triangle is 180°.

Answer:

2. The square root of 49 is 6.

Answer:

3. An algebraic expression can contain variables, numbers, and operations but not **exponents**.

Answer:

4. The area of a circle is calculated as **πr^2**.

Answer:

5. A **ratio** compares two quantities of the same unit.

Answer:

One Word Answer

1. Name a shape with three sides.

Answer:

2. What is the symbol for percentage?

Answer:

3. Name a rational number expressed as a fraction.

Answer:

4. What is the distance around a circle called?

Answer:

5. What is the term for a flip symmetry?

Answer:

Match the Following

Question 1

Column A: Ratio

Column B:

(a) Comparison of quantities

(b) Power of a number

(c) Symmetry

(d) Perimeter

Answer:

Question 2

Column A: Percentage

Column B:

(a) Part of a whole expressed in hundredths

(b) Equal parts of a whole

(c) Distance around a shape

(d) Power of a number

Answer:

Question 3

Column A: Symmetry

Column B:

(a) Identical halves across a line

(b) Rotation by a certain angle

(c) Comparison of quantities

(d) Power of a number

Answer:

Question 4

Column A: Exponent

Column B:

(a) Number of times a base is multiplied by itself

(b) Part of a whole expressed in hundredths

(c) Distance around a shape

(d) Comparison of quantities

Answer:

Question 5

Column A: Perimeter

Column B:

(a) Distance around a shape

(b) Area of a shape

(c) Volume of a solid

(d) Equal parts of a whole

Answer:

Case Based Questions

A school garden is shaped like a rectangle **12 m** long and **8 m** wide, with a square flower bed of side **4 m** inside it, and a circular pond of radius **2 m** at one corner. The remaining area is covered with grass.

1. What is the area of the rectangular garden?

   Answer:

2. Find the area of the square flower bed.

   Answer:

3. Calculate the area of the circular pond (use $\pi \approx 3.14$).

   Answer:

All the Best! ✨